

Worksheet 1 · Spot on facts, pp. 226–227

Technologies that change society

The facts at a glance — your foundation for the Kolloquium and for any factual text on science and technology.

1 Developments in the 21st century

Genetic engineering

CRISPR = tool that *precisely edits* the genetic code of organisms.

- + drought-resistant crops; better treatments for incurable diseases (e.g. stem cells from human embryos)
- morality of the research; risks of human *interference* with DNA

Eco-technology

Ocean clean-ups try to remove the ever-growing amount of plastic waste.

- + solar, hydrogen and wind power are more affordable and efficient; the share of electric vehicles is rising
- fossil-fuel cars still outnumber e-cars in most countries; *raw materials* (lithium, rare-earth metals for batteries) create new environmental problems

AI & digital content

Large language models turn *prompts* into texts, audiovisual content and complex programs.

- + masses of information and data can be re-used in myriad contexts
- often impossible to know where content comes from and how trustworthy it is

2 Artificial intelligence (AI)

Definition: AI is the *simulation of human intelligence in machines* — learning, problem-solving, *reasoning* and perceiving. The field *encompasses* anything from online search engines to military weapons and is progressing rapidly.

Weak / narrow AI

single-task oriented, e.g. personal assistants, video games

Careful: most experts today use *general AI* (AGI) for a still hypothetical AI with human-level intelligence in *all* tasks and count self-driving cars as narrow AI.

→ **Key consequence:** checking data and recognising manipulation matter more than ever — for individuals, companies and governments.

Strong / general AI

more complex, human-like, e.g. (book) self-driving cars, robotic systems

Impact on society

- + benefits everyday life, education and health care
- + testing and scrutiny (e.g. combining research from diverse sources) can reduce *bias*, improve faulty systems and make slow processes far more efficient
- affects human employment
- ethical questions: telling human from AI texts, video *deepfakes*
- manipulation, misinformation, copyright infringements

3 Augmented reality (AR) and virtual reality (VR) — both types of extended reality (XR)

Augmented reality (AR)

What it does *superimposes* interactive digital elements (visual *overlays*, haptic feedback, sensory projections) on the real world → *alters* our perception of it

Devices devices with cameras (e.g. smartphone, tablet)

Used in entertainment, medicine, other commercial industries

Concerns data extracted by XR companies

Virtual reality (VR)

replaces our perception of the real world with a computer-generated 360° environment, fully *immersive* or non-immersive

headset and motion controllers; simulators

gaming, marketing, tourism, education, medicine, *warfare*

may affect a person's perception of reality; data extraction

Our texts: Virtual Helsinki = VR tourism (Baxter, p. 230) · mind uploading = a "virtual afterlife" (Graziano, p. 234)

4 Utopia or dystopia — where will technology take us?

Utopia Greek *ou* (not) + *topos* (place) = "no place"; sounds like *eu* (good) = "good place".

Coined by **Thomas More** in 1516: the perfect island of his book could never really exist.

→ an **imaginary, ideal society**; by highlighting aspects of a better world it *criticises contemporary society* and suggests changes.

Dystopia Greek *dys-* (bad) + *topos*.

Also political: a **critique of contemporary socio-political realities** that shows how bad things *could* become — worst-case scenarios if issues stay unaddressed.

→ often set in the **not-too-distant future**, in societies readers recognise; themes: politics, economics, identity, privacy, freedom, health, science and technology.

